GAUTHAM ANNE

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EDUCATION

BS Mechanical Engineering, MS Electrical Engineering, Northwestern University, GPA: 3.76 2023-2027
Relevant Courses: Real-Time Design and Verification with FPGAs, Robotic Manipulation, CMOS VLSI Circuits Design, Machine Learning, Heat Transfer, DSP, Nonlinear Control, Nanophotonics, Vibrations, Advanced Electrodynamics, Random Processes, Dynamical Systems, Feedback Systems, Fluid Mechanics, Thermodynamics

High School Diploma, Illinois Mathematics and Science Academy GPA: 3.96 2020-2023
Student Researcher in HEPG at Fermilab — Robotics Team Captain — 3x AIME Qualifier — Editor-in-Chief of Newspaper — Author — Junior Counselor at Ross Mathematics Program — TKD Sparring Team (ITSO State 4th Place)

SKILLS

Circuits & AMS	Verilog, SystemVerilog, SPICE (LTspice, Cadence Spectre), PLL/ADC/DAC
FPGA & RTL Design	FPGA prototyping (Xilinx, Intel, Gowin), SystemVerilog testbenches, fixed-point DSP
PCB & CAD Tools	Cadence Virtuoso, Altium, KiCad, EAGLE, SolidWorks, Cadence Allegro
Programming & Simulation	MATLAB/Simulink, Python, C/C++, Microchip Studio, STM32CubeIDE
Machine Learning & Data	TensorFlow, Keras, scikit-learn, NumPy, pandas, Flask

EXPERIENCE

Project Manager, IEEE Student Branch — Northwestern University Nov 2025 – Present
Custom DLP-based maskless lithography platform for sub-micron patterning *Evanston, IL*

Teaching Assistant - Northwestern University Jan 2025 - Present
Engineering Analysis III (Spring 2025), Engineering Analysis I (Fall 2025, Winter 2026) *Evanston, IL*

Northwestern Haptics Group (advised by Professors Colgate & Peshkin) Sept 2023 - Present
Undergraduate Researcher in Haptics Development *Evanston, IL*

- Designed and built a high-impedance haptic probe for fingertip impedance measurement,
- Integrated voice-coil based actuation, LDV velocity sensing, and current-derived force estimation.
- Performed frequency-domain impedance extraction using broadband chirp excitation to characterize stiffness, damping, and inertial dynamics across touch-relevant frequencies (30–300Hz).

MIT Quantum & Precision Measurements Group (advised by Professor Sudhir) Jun 2024 - Dec 2024
Visting Scholar, Electrical Network Theory Research *Cambridge, MA*

- Studied theory for optimizing circuit synthesis (multiport synthesis methods) by minimizing Nyquist noise
- Built Mathematica/SPICE framework for node-level thermal noise analysis; explored impedance-matching for optimal SNR

Omnid Research Group (advised by Professors Elwin & Lynch) Oct 2023 - Feb 2024
MARS Omnid Team *Evanston, IL*

- Prepared Omnid Mocobots, collaborative mobile manipulators, for the 2024 Amazon MARS conference
- Upgraded JC satellite boards (Tiva Launchpad), designed PCB shielding, and implemented E-Stop recovery via STOs

Dave's Italian Kitchen Sept 2023 - Present
Hosting, Waiting, & Dishwashing *Evanston, IL*

Fermi National Accelerator Laboratory (advised by Dr. Dong) Aug 2021 - Jun 2023
Undergraduate Researcher in High Energy Physics *Batavia, IL*

- Built DL neural nets for prompt lepton jet classification; Applied multivariate cuts to suppress backgrounds in H^{++} dilepton searches; results presented at APS April Meeting 2023

PROJECTS

Low-Cost Scanning Tunneling Microscop Designed a functional STM, from scratch, to image materials at atomic resolution. Built low noise regulated linear power supply, a tunneling amplifier, lock-in amplifier, unimorph disk scanner piezo driver, & awarded "Most Technical" & "Most Creative" in EE327 Design Showcase.

Other Projects Lock-In Amplifier AMS SoC (Sky130), FPGA-Based Canny Edge Extraction Pipeline, Custom Webcam PCB, Full fledged DC motor PID controller, Jack-in-Box Lagrangian Mech Simulation, Low-cost EEG. See my portfolio.